

ICDFA FELLOWSHIP

International Centre for Digital Forensics & Analytics

WADF-2026-M03 | Network Security

Week 09

Student Laboratory Report

Network & Service Reconnaissance — Nmap Fundamentals, Service Enumeration, Safe NSE Scripts and WhatWeb Fingerprinting



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Fellowship Program	ICDFA Fellowship in Cybersecurity & Digital Forensics
Module	WADF-2026-M03 — Network Security
Week / Lab Covered	Week 09 — Lab 01: Network Service Reconnaissance
Attacker Machine	Kali Linux (rolling release) — 192.168.19.131
Target Machine	Metasploitable 2 — 192.168.19.133
Date of Submission	09/06/2026

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Declaration: The work presented in this report was performed by me on my own Kali Linux virtual machine, and all screenshots are my own captures.

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Authorisation Boundary

I used only the authorised Metasploitable 2 VM as the target. No public systems, random IP addresses, or production infrastructure were scanned.

1. Introduction

1.1 Purpose

This report documents the network and service reconnaissance conducted against the Metasploitable 2 virtual machine as part of Lab 1. The objective was to identify the target, perform various Nmap scans, enumerate services, and fingerprint web technologies using WhatWeb.

1.2 Lab Environment

The lab consisted of two virtual machines:

- Kali Linux: Attacker machine (192.168.19.131)
- Metasploitable 2: Target machine (192.168.19.133)

1.3 Authorisation Boundary

All scans were performed exclusively within the authorised lab environment. No public systems, random IP addresses, or production infrastructure were scanned.

2. Lab Preparation

2.1 Find Kali IP

Objective: Identify the Kali Linux IP address.

```
$ ip addr
```

Theory: The `ip addr` command displays all network interfaces and their IP addresses. The active Ethernet interface (`eth0`) contains the lab IP address. The loopback interface (`127.0.0.1`) should be ignored as it's localhost.

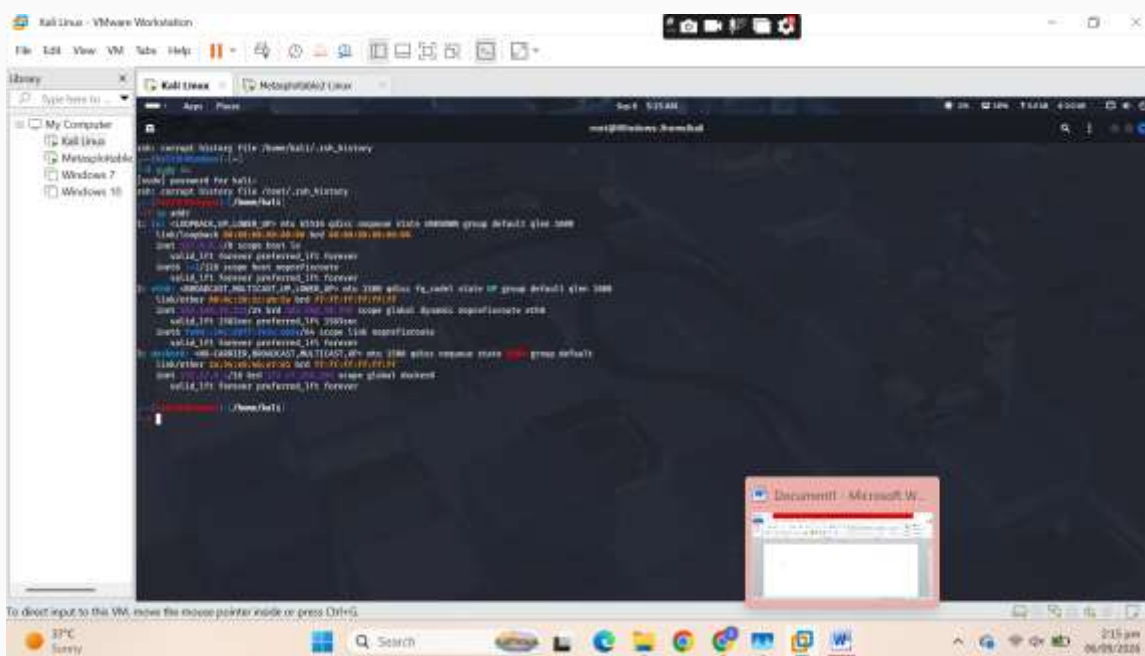


Figure 1: Find Kali IP

2.2 Find Metasploitable 2 IP

Objective: Identify the authorised target IP address.

```
$ ifconfig
```

Theory: The ifconfig command displays network interface configuration. On Metasploitable 2, the eth0 interface shows the target IP address.

```

up
sudo! password for msfadmin:
orry, try again.
sudo! password for msfadmin:
sfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:21:2e:2e
          inet addr:192.168.19.101  Bcast:192.168.19.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe21:2e2e/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:22057 errors:39 dropped:0 overruns:0 frame:0
          TX packets:463 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1499239 (1.4 MB)  TX bytes:29940 (29.2 KB)
          Interrupt:17 Base address:0x2000

          Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:684 errors:0 dropped:0 overruns:0 frame:0
          TX packets:684 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:171133 (167.1 KB)  TX bytes:171133 (167.1 KB)

sfadmin@metasploitable:~$ _

```

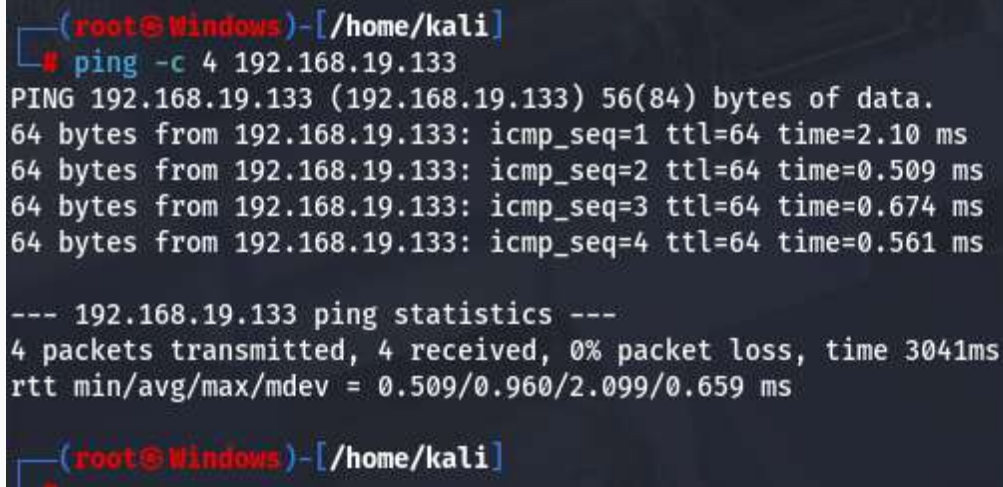
Figure 2: Find Metasploitable 2 IP

2.3 Confirm Connectivity

Objective: Verify the VMs can communicate.

```
$ ping -c 4 192.168.19.133
```

Theory: The ping command sends ICMP echo requests to test network connectivity. The -c 4 option limits the test to 4 packets. Successful replies confirm that Kali and Metasploitable can communicate.



```
(root@Windows)-[/home/kali]
# ping -c 4 192.168.19.133
PING 192.168.19.133 (192.168.19.133) 56(84) bytes of data.
64 bytes from 192.168.19.133: icmp_seq=1 ttl=64 time=2.10 ms
64 bytes from 192.168.19.133: icmp_seq=2 ttl=64 time=0.509 ms
64 bytes from 192.168.19.133: icmp_seq=3 ttl=64 time=0.674 ms
64 bytes from 192.168.19.133: icmp_seq=4 ttl=64 time=0.561 ms

--- 192.168.19.133 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3041ms
rtt min/avg/max/mdev = 0.509/0.960/2.099/0.659 ms

(root@Windows)-[/home/kali]
```

Figure 3: Confirm Connectivity

3. Core Nmap Scans

3.1 Host Discovery

Objective: Discover live hosts without a normal port scan.

```
$ nmap -sn 192.168.19.0/24
```

Theory: The `-sn` flag performs a ping sweep (host discovery) without scanning ports. This identifies all active hosts on the network.

Results: The scan found 5 live hosts including Kali (192.168.19.131) and Metasploitable (192.168.19.133).

```
(root@Windows)-[/home/kali]
$ nmap -sn 192.168.19.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 06:36 EDT
Nmap scan report for 192.168.19.1
Host is up (0.00086s latency).
MAC Address: 00:50:56:C0:00:08 (VMware)
Nmap scan report for 192.168.19.2
Host is up (0.00024s latency).
MAC Address: 00:50:56:E2:2B:94 (VMware)
Nmap scan report for 192.168.19.133
Host is up (0.00049s latency).
MAC Address: 00:0C:29:21:2E:2E (VMware)
Nmap scan report for 192.168.19.254
Host is up (0.00021s latency).
MAC Address: 00:50:56:F4:22:1E (VMware)
Nmap scan report for 192.168.19.131
Host is up.
Nmap done: 256 IP addresses (5 hosts up) scanned in 2.00 seconds
(root@Windows)-[/home/kali]
```

Figure 4: Host Discovery

3.2 Default TCP Scan

Objective: Find commonly scanned open TCP ports.

```
$ nmap 192.168.19.133
```

Theory: Nmap's default scan checks the top 1,000 most common TCP ports. It displays open ports, their states, and basic service identification.

Results: Metasploitable revealed 23 open ports including FTP, SSH, HTTP, and MySQL.

```
(root@Windows)-[/home/kali]
# nmap 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 06:37 EDT
Nmap scan report for 192.168.19.133
Host is up (0.0019s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:21:2E:2E (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.37 seconds
```

Figure 5: Default TCP Scan

3.3 Service/Version Detection

Objective: Identify software behind open ports.

```
$ nmap -sV 192.168.19.133
```

Theory: The `-sV` flag enables service/version detection. Nmap sends service probes to identify specific software products and version numbers.


```

(root@Windows)-[/home/kali]
$ nmap -sV 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 06:37 EDT
Nmap scan report for 192.168.19.133
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login?
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:21:2E:2E (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux;

Service detection performed. Please report any incorrect results at https://nmap.org/submit/
Nmap done: 1 IP address (1 host up) scanned in 11.88 seconds

(root@Windows)-[/home/kali]
$

```

Figure 6: Service/Version Detection

3.4 Maximum Version Intensity

Objective: Understand deeper version probing.

```
$ nmap -sV --version-intensity 9 192.168.19.133
```

Theory: Intensity 9 tells Nmap to use its full version-probe set, attempting to extract the maximum possible version information.

3.5 OS Detection

Objective: Estimate the target operating system.

```
$ sudo nmap -O 192.168.19.133
```

Theory: The -O flag enables OS fingerprinting, analyzing TCP/IP stack characteristics to determine the operating system. Root privileges are required for optimal results.

Results: The scan identified Linux kernel 2.6.9-2.6.33.

```
(root@Windows)-[/home/kali]
$ sudo nmap -O 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 06:39 EDT
Nmap scan report for 192.168.19.133
Host is up (0.00086s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 00:0C:29:21:2E:2E (VMware)
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6
OS details: Linux 2.6.9 - 2.6.33
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.73 seconds
```

Figure 8: OS Detection

3.6 Aggressive Scan

Objective: Combine several discovery functions.

```
$ sudo nmap -A 192.168.19.133
```

Theory: The -A flag combines OS detection, version detection, default NSE scripts, and traceroute. It's noisier because it makes multiple probes and connections.

Results: The aggressive scan provided comprehensive information including anonymous FTP access, SSH host keys, SSL certificates, SMB details, and VNC protocol information.

```

root@kali:~/home/kali# sudo nmap -A 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 06:40 EDT
Nmap scan report for 192.168.19.133
Host is up (0.00065s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|_STAT:
|_FTP server status:
|_  Connected to 192.168.19.131
|_  Logged in as ftp
|_  TYPE: ASCII
|_  No session bandwidth limit
|_  Session timeout in seconds is 300
|_  Control connection is plain text
|_  Data connections will be plain text
|_  vsFTPd 2.3.4 - secure, fast, stable
|_End of status
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
|_ssh-hostkey:
|_ 1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_ 2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=DCOSA/stateOrProvinceName=There is no
such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES,
8BITMIME, DSN
|_sslv2:
|_SSLv2 supported
|_ciphers:
|_  SSL2_RC4_128_EXPORT40_WITH_MD5
|_  SSL2_RC2_128_CBC_EXPORT40_WITH_MD5
|_  SSL2_DES_192_EDE3_CBC_WITH_MD5
|_  SSL2_DES_64_CBC_WITH_MD5
|_  SSL2_RC2_128_CBC_WITH_MD5

```

Figure 9: Aggressive Scan

3.7 Full TCP Port Scan

Objective: Check all 65,535 ports.

```
$ sudo nmap -p- 192.168.19.133
```

Theory: The -p- flag scans all TCP ports from 1 to 65,535. This is more thorough than the default scan.

Results: Revealed additional ports including high-numbered ports (32901, 38448, 46947, 59707) which are RPC services.

```

$ sudo nmap -p- 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 06:42 EDT
Nmap scan report for 192.168.19.133
Host is up (0.0012s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
32901/tcp open  unknown
38448/tcp open  unknown
46947/tcp open  unknown
59707/tcp open  unknown
MAC Address: 00:0C:29:21:2E:2E (VMware)

Nmap done: 1 IP address (1 host up) scanned in 6.05 seconds

```

Figure 10: Full TCP Port Scan

3.8 Full TCP + Versions

Objective: Build a fuller service inventory.

```
$ sudo nmap -p- -sV 192.168.19.133
```

Theory: Combining -p- with -sV provides complete port coverage with version information.

Results: Revealed all 31 open ports with their specific versions, including distccd, PostgreSQL, and Ruby DRb services.


```

kali@kali:~/home/kali$ sudo nmap -p- -sV 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 06:43 EDT
Nmap scan report for 192.168.19.133
Host is up (0.0014s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
53/tcp    open  domain       ISC BIND 9.4.2
80/tcp    open  http         Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind      2 (RPC #100000)
139/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec         netkit-rsh rexecd
513/tcp   open  login        OpenBSD or Solaris rlogind
514/tcp   open  tcpwrapped
1099/tcp  open  java-rmi     GNU Classpath grmiregistry
1524/tcp  open  bindshell    Metasploitable root shell
2049/tcp  open  nfs          2-4 (RPC #100003)
2121/tcp  open  ftp          ProFTPD 1.3.1
3306/tcp  open  mysql        MySQL 5.0.51a-3ubuntu5
3632/tcp  open  distccd      distccd v1 ((GNU) 4.2.4 (Ubuntu 4.2.4-1ubuntu4))
5432/tcp  open  postgresql   PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc          VNC (protocol 3.3)
6000/tcp  open  X11          (access denied)
6667/tcp  open  irc          UnrealIRCd
6697/tcp  open  irc          UnrealIRCd
8009/tcp  open  ajp13        Apache Jserv (Protocol v1.3)
8180/tcp  open  http         Apache Tomcat/Coyote JSP engine 1.1
8787/tcp  open  drb          Ruby DRB RMI (Ruby 1.8; path /usr/lib/ruby/1.8/drbc)
32901/tcp open  mountd       1-3 (RPC #100005)
38448/tcp open  status       1 (RPC #100024)
46947/tcp open  nlockmgr     1-4 (RPC #100021)
59707/tcp open  java-rmi     GNU Classpath grmiregistry
MAC Address: 00:0C:29:21:2E:2E (VMware)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

```

Figure 11: Full TCP + Versions

3.9 Timing Template

Objective: See how a local-lab scan can be accelerated.

```
$ sudo nmap -p- -T4 192.168.19.133
```

Theory: The -T4 flag uses a faster timing template suited to reliable isolated lab networks.

Results: The scan completed in approximately 5.93 seconds, demonstrating improved performance compared to the default timing.

```

$ sudo nmap -p- -T4 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 06:47 EDT
Nmap scan report for 192.168.19.133
Host is up (0.0014s latency).
Not shown: 65505 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
3632/tcp  open  distccd
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
6697/tcp  open  ircs-u
8009/tcp  open  ajp13
8180/tcp  open  unknown
8787/tcp  open  msgsrvr
32901/tcp open  unknown
38448/tcp open  unknown
46947/tcp open  unknown
59707/tcp open  unknown
MAC Address: 00:0C:29:21:2E:2E (VMware)

Nmap done: 1 IP address (1 host up) scanned in 5.93 seconds

```

Figure 12: Timing Template

3.10 Selected Ports

Objective: Focus on known services.

```
$ nmap -p 21,22,23,25,80,139,445 192.168.19.133
```

Theory: The -p flag with comma-separated ports scans only specified ports. This focused scanning is useful when you want to check specific services quickly.

Results: The scan completed in just 0.27 seconds.

```
(root@Windows)-[/home/kali]
# nmap -p 21,22,23,25,80,139,445 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 06:48 EDT
Nmap scan report for 192.168.19.133
Host is up (0.00044s latency).

PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
80/tcp    open  http
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
MAC Address: 00:0C:29:21:2E:2E (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.27 seconds
```

Figure 13: Selected Ports

3.11 Port Range

Objective: Scan a continuous range.

```
$ nmap -p 1-1024 192.168.19.133
```

Theory: A hyphen defines an inclusive port range. Scanning ports 1 through 1024 (well-known ports) reveals common services.

```

(root@Windows)-[/home/kali]
# nmap -p 1-1024 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 06:48 EDT
Nmap scan report for 192.168.19.133
Host is up (0.0028s latency).
Not shown: 1012 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
MAC Address: 00:0C:29:21:2E:2E (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.35 seconds

```

Figure 14: Port Range

3.12 UDP Scan

Objective: Discover UDP services.

```
$ sudo nmap -sU 192.168.19.133
```

Theory: UDP scans take longer because UDP is connectionless and doesn't respond to closed ports reliably. Nmap must wait for timeouts.

Results: Identified UDP services including DNS (53), DHCP (68), TFTP (69), rpcbind (111), NetBIOS (137, 138), and NFS (2049).


```

(root@Windows)-[/home/kali]
# sudo nmap -sU 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 06:49 EDT
Stats: 0:04:49 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 28.98% done; ETC: 07:05 (0:11:48 remaining)
Stats: 0:05:56 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 35.68% done; ETC: 07:05 (0:10:42 remaining)
Stats: 0:07:59 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 46.98% done; ETC: 07:06 (0:09:01 remaining)
Stats: 0:10:22 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 61.28% done; ETC: 07:06 (0:06:33 remaining)
Stats: 0:13:18 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 78.78% done; ETC: 07:06 (0:03:35 remaining)
Stats: 0:14:12 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 84.18% done; ETC: 07:06 (0:02:40 remaining)
Stats: 0:15:25 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 91.48% done; ETC: 07:06 (0:01:26 remaining)
Stats: 0:16:27 elapsed; 0 hosts completed (1 up), 1 undergoing UDP Scan
UDP Scan Timing: About 96.68% done; ETC: 07:06 (0:00:34 remaining)
Nmap scan report for 192.168.19.133
Host is up (0.00063s latency).
Not shown: 993 closed udp ports (port-unreach)
PORT      STATE      SERVICE
53/udp    open       domain
68/udp    open|filtered dhcpc
69/udp    open|filtered tftp
111/udp   open       rpcbind
137/udp   open       netbios-ns
138/udp   open|filtered netbios-dgm
2049/udp  open       nfs
MAC Address: 00:0C:29:21:2E:2E (VMware)

Nmap done: 1 IP address (1 host up) scanned in 1029.59 seconds
(root@Windows)-[/home/kali]

```

Figure 15: UDP Scan

3.13 Top UDP Ports + Versions

Objective: Keep UDP practical for beginners.

```
$ sudo nmap -sU --top-ports 20 -sV 192.168.19.133
```

Theory: Scanning only the top 20 UDP ports with version detection is more practical.

Results: Confirmed ISC BIND 9.4.2 on port 53 and NetBIOS service on port 137. Several ports showed "open|filtered" status.

```

root@kali:~# sudo nmap -sU --top-ports 20 -sV 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 07:07 EDT
Stats: 0:00:16 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 16.67% done; ETC: 07:08 (0:00:50 remaining)
Nmap scan report for 192.168.19.133
Host is up (0.00057s latency).

PORT      STATE      SERVICE      VERSION
53/udp    open       domain       ISC BIND 9.4.2
67/udp    closed
68/udp    open|filtered dhcpc
69/udp    open|filtered tftp
123/udp   open|filtered ntp
135/udp   open|filtered msrpc
137/udp   open       netbios-ns   Microsoft Windows netbios-ns (workgroup: WORKGROUP)
138/udp   open|filtered netbios-dgm
139/udp   closed     netbios-ssn
161/udp   closed     snmp
162/udp   open|filtered snmptrap
445/udp   closed     microsoft-ds
500/udp   closed     isakmp
514/udp   open|filtered syslog
520/udp   closed     route
631/udp   open|filtered ipp
1434/udp  open|filtered ms-sql-m
1900/udp  closed     upnp
4500/udp  open|filtered nat-t-ike
49152/udp closed     unknown
MAC Address: 00:0C:29:21:2E:2E (VMware)
Service Info: Host: METASPLOITABLE; OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 110.81 seconds

```

Figure 16: Top UDP Ports + Versions

4. Safe Nmap NSE Enumeration

4.1 Default Scripts

Objective: Collect common service information.

```
$ nmap -sC -sV 192.168.19.133
```

Theory: The `-sC` flag runs Nmap's default NSE scripts for detailed service enumeration. Combined with `-sV`, this provides comprehensive findings.

Results: Found anonymous FTP login allowed, SSH host keys, SSL certificates, DNS version, SMB security information, and VNC protocol details.

```

root@kali:~/home/kali# nmap -sC -sV 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 07:10 EDT
Nmap scan report for 192.168.19.133
Host is up (0.0021s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
| ftp-syst:
|   STAT:
|   FTP server status:
|     Connected to 192.168.19.131
|     Logged in as ftp
|     TYPE: ASCII
|     No session bandwidth limit
|     Session timeout in seconds is 300
|     Control connection is plain text
|     Data connections will be plain text
|     vsFTPd 2.3.4 - secure, fast, stable
|_End of status
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
22/tcp    open  ssh          OpenSSH 4.7p1 Debian Subuntu1 (protocol 2.0)
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|   2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
23/tcp    open  telnet       Linux telnetd
25/tcp    open  smtp         Postfix smtpd
|_ssl-date: 2026-09-06T11:10:46+00:00; +6s from scanner time.
|_ssl-cert: Subject: commonName=ubuntu804-base.localdomain/organizationName=OCOSA/stateOrProvinceName=There is no
such thing outside US/countryName=XX
|_Not valid before: 2010-03-17T14:07:45
|_Not valid after: 2010-04-16T14:07:45
|_smtp-commands: metasploitable.localdomain, PIPELINING, SIZE 10240000, VRFY, ETRN, STARTTLS, ENHANCEDSTATUSCODES,
8BITMIME, DSN
|_sslv2:
|   SSLv2 supported
|   ciphers:
|     SSL2_RC2_128_CBC_WITH_MD5
|     SSL2_DES_192_EDE3_CBC_WITH_MD5

```

Figure 17: Default Scripts

4.2 HTTP Title

Objective: Read the title of the lab web page.

```
$ nmap -p 80 --script http-title 192.168.19.133
```

Theory: The http-title script requests the HTML page and extracts the title tag.

Results: The result shows "Metasploitable2 - Linux", confirming the target is the Metasploitable 2 VM.

```
(root@Windows)-[/home/kali]
# nmap -p 80 --script http-title 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 07:13 EDT
Nmap scan report for 192.168.19.133
Host is up (0.00036s latency).

PORT      STATE SERVICE
80/tcp    open  http
|_http-title: Metasploitable2 - Linux
MAC Address: 00:0C:29:21:2E:2E (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.29 seconds

(root@Windows)-[/home/kali]
```

Figure 18: HTTP Title

4.3 HTTP Headers

Objective: Inspect response metadata.

```
$ nmap -p 80 --script http-headers 192.168.19.133
```

Theory: The http-headers script displays HTTP response headers.

Results: Revealed Server: Apache/2.2.8 (Ubuntu) DAV/2 and X-Powered-By: PHP/5.2.4-2ubuntu5.10.

```

(root@Windows)-[/home/kali]
# nmap -p 80 --script http-headers 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 07:13 EDT
Nmap scan report for 192.168.19.133
Host is up (0.00043s latency).

PORT      STATE SERVICE
80/tcp    open  http
| http-headers:
|   Date: Sun, 06 Sep 2026 11:13:30 GMT
|   Server: Apache/2.2.8 (Ubuntu) DAV/2
|   X-Powered-By: PHP/5.2.4-2ubuntu5.10
|   Connection: close
|   Content-Type: text/html
|
|_ (Request type: HEAD)
MAC Address: 00:0C:29:21:2E:2E (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.27 seconds

```

Figure 19: HTTP Headers

4.4 SMB Protocols

Objective: Inspect supported SMB protocol versions.

```
$ nmap -p 139,445 --script smb-protocols 192.168.19.133
```

Theory: The smb-protocols script queries the SMB service for protocol support.

Results: Revealed NT LM 0.12 (SMBv1), which is considered dangerous as it has known vulnerabilities.

```
(root@Windows)-[/home/kali]
# nmap -p 139,445 --script smb-protocols 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 07:13 EDT
Nmap scan report for 192.168.19.133
Host is up (0.00034s latency).

PORT      STATE SERVICE
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
MAC Address: 00:0C:29:21:2E:2E (VMware)

Host script results:
| smb-protocols:
|   dialects:
|_    NT LM 0.12 (SMBv1) [dangerous, but default]

Nmap done: 1 IP address (1 host up) scanned in 0.30 seconds
```

Figure 20: SMB Protocols

4.5 SSH Host Keys

Objective: Observe SSH server identity fingerprints.

```
$ nmap -p 22 --script ssh-hostkey 192.168.19.133
```

Theory: SSH servers publish host keys for client verification.

Results: Identified DSA (1024-bit) and RSA (2048-bit) keys.


```
(root@Windows)-[/home/kali]
# nmap -p 22 --script ssh-hostkey 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 07:14 EDT
Nmap scan report for 192.168.19.133
Host is up (0.00033s latency).

PORT      STATE SERVICE
22/tcp    open  ssh
| ssh-hostkey:
|   1024 60:0f:cf:e1:c0:5f:6a:74:d6:90:24:fa:c4:d5:6c:cd (DSA)
|_  2048 56:56:24:0f:21:1d:de:a7:2b:ae:61:b1:24:3d:e8:f3 (RSA)
MAC Address: 00:0C:29:21:2E:2E (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.32 seconds
```

Figure 21: SSH Host Keys

4.6 FTP Banner

Objective: Compare a banner with -sV output.

```
$ nmap -p 21 -sV --script banner 192.168.19.133
```

Theory: The banner script reads service banners.

Results: The FTP banner displayed "220 (vsFTPD 2.3.4)", confirming the same version identified by -sV.

```
(root@Windows)-[/home/kali]
# nmap -p 21 -sV --script banner 192.168.19.133
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 07:14 EDT
Nmap scan report for 192.168.19.133
Host is up (0.00070s latency).

PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 2.3.4
|_ banner: 220 (vsFTPD 2.3.4)
MAC Address: 00:0C:29:21:2E:2E (VMware)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.36 seconds
```

Figure 22: FTP Banner

5. Manual HTTP Confirmation

5.1 Request Headers with curl

Objective: Confirm what the web service returns without a vulnerability scanner.

```
$ curl -I http://192.168.19.133
```

Theory: The curl -I command requests HTTP headers only. This manually confirms the web server information.

Results: Matched Nmap findings: HTTP/1.1 200 OK, Server: Apache/2.2.8 (Ubuntu) DAV/2, and X-Powered-By: PHP/5.2.4-2ubuntu5.10.

A terminal window screenshot showing the command 'curl -I http://192.168.19.133' being executed. The prompt is '(root@Windows)-[/home/kali]'. The output of the command is displayed in white text on a dark background: 'HTTP/1.1 200 OK', 'Date: Sun, 06 Sep 2026 11:15:20 GMT', 'Server: Apache/2.2.8 (Ubuntu) DAV/2', 'X-Powered-By: PHP/5.2.4-2ubuntu5.10', and 'Content-Type: text/html'.

Figure 23: Request Headers with curl

6. WhatWeb Fingerprinting

6.1 Basic Fingerprint

Objective: Identify exposed web technologies.

```
$ whatweb http://192.168.19.133
```

Theory: WhatWeb performs web technology fingerprinting. The default scan identifies technologies including HTTP servers, frameworks, and CMS.

Results: Identified Apache 2.2.8, PHP 5.2.4-2ubuntu5.10, WebDAV, and the page title.

```

root@kali:~/home/kali# whatweb http://192.168.19.133
http://192.168.19.133 [200 OK] Apache[2.2.8], Country[RESERVED][22], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.19.133], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

```

Figure 24: Basic Fingerprint

6.2 Verbose Output

Objective: See plugin/fingerprint detail.

```
$ whatweb -v http://192.168.19.133
```

Theory: The -v flag shows verbose information including plugin details, descriptions, and complete HTTP headers.

```

root@kali:~/home/kali# whatweb http://192.168.19.133
http://192.168.19.133 [200 OK] Apache[2.2.8], Country[RESERVED][22], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.19.133], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

root@kali:~/home/kali# whatweb -v http://192.168.19.133
WhatWeb report for http://192.168.19.133:
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.19.133
Country     : RESERVED, 22

Summary     : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
  The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.

  Version      : 2.2.8 (from HTTP Server Header)
  Google Dorks : (3)
  Website      : http://httpd.apache.org/

[ HTTPServer ]
  HTTP server header string. This plugin also attempts to identify the operating system from the server header.

  OS           : Ubuntu Linux
  String       : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ PHP ]
  PHP is a widely-used general-purpose scripting language that is especially suited for Web development and can be embedded into HTML. This plugin identifies PHP errors, modules and versions and extracts the local file path and

```

Figure 25: Verbose Output

6.3 Aggression Level 1

Objective: Understand the lightest fingerprinting mode.

```
$ whatweb -a 1 http://192.168.19.133
```

Theory: Aggression level 1 keeps requests minimal. The fingerprinting results are basic but sufficient for identifying core web technologies.



```
(root@ kali) ~ [ /home/kali ]
$ whatweb -a 1 http://192.168.19.133
http://192.168.19.133 [200 OK] Apache[2.2.8], Country[RESERVED][22], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.19.133], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]
```

Figure 26: Aggression Level 1

6.4 Aggression Level 3

Objective: Use additional requests in the isolated lab.

```
$ whatweb -a 3 -v http://192.168.19.133
```

Theory: Level 3 makes more requests to improve identification. It provides similar results to level 1 but with slightly more plugin information and detail.

```

root@kali:~/home/kali# whatweb -a 1 http://192.168.19.133
http://192.168.19.133 [200 OK] Apache[2.2.8], Country[RESERVED][22], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[192.168.19.133], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

root@kali:~/home/kali# whatweb -a 3 -v http://192.168.19.133
WhatWeb report for http://192.168.19.133
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.19.133
Country     : RESERVED, 22

Summary    : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]

Detected Plugins:
[ Apache ]
  The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.

  Version      : 2.2.8 (from HTTP Server Header)
  Google Dorks : (3)
  Website      : http://httpd.apache.org/

[ HTTPServer ]
  HTTP server header string. This plugin also attempts to identify the operating system from the server header.

  OS           : Ubuntu Linux
  String       : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ PHP ]
  PHP is a widely-used general-purpose scripting language that is especially suited for Web development and can be embedded into HTML. This plugin identifies PHP errors,

```

Figure 27: Aggression Level 3

6.5 Aggression Level 4

Objective: Observe the heaviest WhatWeb mode.

```
$ whatweb -a 4 -v http://192.168.19.133
```

Theory: Level 4 generates substantially more requests and can take much longer. Level 4 is optional and did not provide significantly more useful information than levels 1 and 3 for this target.

Note: Scan was interrupted after taking too long. Levels 1 and 3 provided sufficient fingerprinting for this lab.

6.6 Follow Redirects

Objective: Fingerprint the destination of redirects.

```
$ whatweb --follow-redirect=always http://192.168.19.133
```

Theory: The --follow-redirect flag follows HTTP redirect responses.

Results: Since the Metasploitable 2 web server responded directly with a 200 OK status, no redirect occurred.



```
(root@kali) ~/home/kali
$ whatweb --follow-redirect=always http://192.168.19.133
http://192.168.19.133 [200 OK] Apache[2.2.8], Country[RESERVED][22], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu
) DAV/2], IP[192.168.19.133], PHP[5.2.4-2ubuntu5.10], Title[Metasploitable2 - Linux], WebDAV[2], X-Powered-By[PHP/
5.2.4-2ubuntu5.10]

(root@kali) ~/home/kali
```

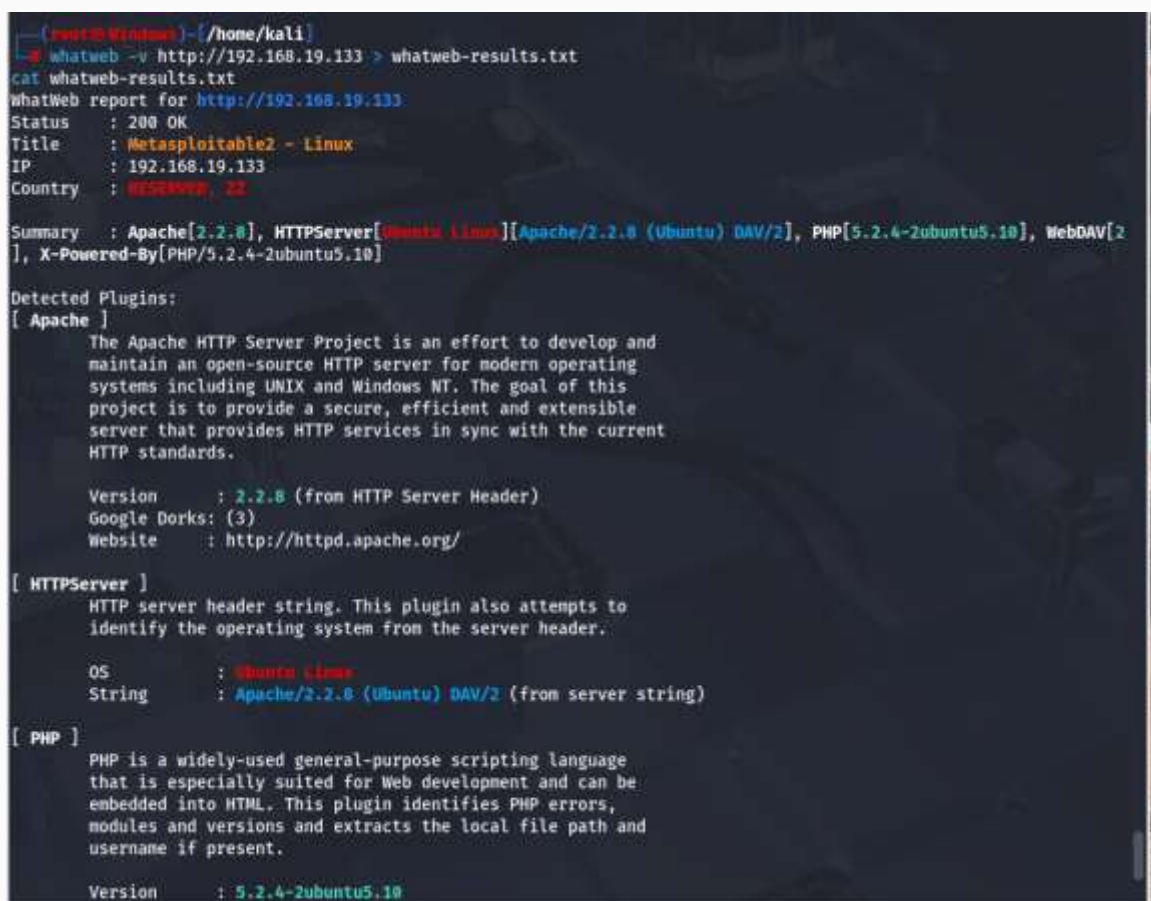
Figure 28: Follow Redirects

6.7 Save Results

Objective: Preserve evidence.

```
$ whatweb -v http://192.168.19.133 > whatweb-results.txt
$ cat whatweb-results.txt
```

Theory: The > operator redirects output to a file. The cat command displays the file contents.



```
(root@kali) ~/home/kali
$ whatweb -v http://192.168.19.133 > whatweb-results.txt
cat whatweb-results.txt
WhatWeb report for http://192.168.19.133
Status      : 200 OK
Title       : Metasploitable2 - Linux
IP          : 192.168.19.133
Country     : RESERVED, 22
Summary     : Apache[2.2.8], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], PHP[5.2.4-2ubuntu5.10], WebDAV[2], X-Powered-By[PHP/5.2.4-2ubuntu5.10]
Detected Plugins:
[ Apache ]
  The Apache HTTP Server Project is an effort to develop and maintain an open-source HTTP server for modern operating systems including UNIX and Windows NT. The goal of this project is to provide a secure, efficient and extensible server that provides HTTP services in sync with the current HTTP standards.

  Version      : 2.2.8 (from HTTP Server Header)
  Google Dorks : (3)
  Website      : http://httpd.apache.org/

[ HTTPServer ]
  HTTP server header string. This plugin also attempts to identify the operating system from the server header.

  OS           : Ubuntu Linux
  String       : Apache/2.2.8 (Ubuntu) DAV/2 (from server string)

[ PHP ]
  PHP is a widely-used general-purpose scripting language that is especially suited for Web development and can be embedded into HTML. This plugin identifies PHP errors, modules and versions and extracts the local file path and username if present.

  Version      : 5.2.4-2ubuntu5.10
```


Figure 29: Save Results

7. Service Inventory

Port	Proto	State	Service	Version / Evidence	Command
21/tcp	TCP	open	ftp	vsftpd 2.3.4	nmap -sV
22/tcp	TCP	open	ssh	OpenSSH 4.7p1 Debian 8ubuntu1	nmap -sV
23/tcp	TCP	open	telnet	Linux telnetd	nmap -sV
25/tcp	TCP	open	smtp	Postfix smtpd	nmap -sV
53/tcp	TCP	open	domain	ISC BIND 9.4.2	nmap -sV
80/tcp	TCP	open	http	Apache httpd 2.2.8 (Ubuntu) DAV/2	nmap -sV
111/tcp	TCP	open	rpcbind	2 (RPC #100000)	nmap -sV
139/tcp	TCP	open	netbios-ssn	Samba smbd 3.X - 4.X	nmap -sV
445/tcp	TCP	open	netbios-ssn	Samba smbd 3.0.20-Debian	nmap -sV
512/tcp	TCP	open	exec	netkit-rsh rexecd	nmap -sV
513/tcp	TCP	open	login	OpenBSD or Solaris rlogind	nmap -sV
514/tcp	TCP	open	tcpwrapped	tcpwrapped	nmap -sV
1099/tcp	TCP	open	java-rmi	GNU Classpath grmiregistry	nmap -sV
1524/tcp	TCP	open	bindshell	Metasploitable root shell	nmap -sV
2049/tcp	TCP	open	nfs	2-4 (RPC #100003)	nmap -sV
2121/tcp	TCP	open	ftp	ProFTPD 1.3.1	nmap -sV
3306/tcp	TCP	open	mysql	MySQL 5.0.51a-3ubuntu5	nmap -sV
3632/tcp	TCP	open	distccd	distccd v1 (GNU) 4.2.4	nmap -sV
5432/tcp	TCP	open	postgresql	PostgreSQL 8.3.0 - 8.3.7	nmap -sV
5900/tcp	TCP	open	vnc	VNC protocol 3.3	nmap -sV
6000/tcp	TCP	open	X11	access denied	nmap -sV
6667/tcp	TCP	open	irc	UnrealIRCd	nmap -sV
6697/tcp	TCP	open	irc	UnrealIRCd	nmap -sV
8009/tcp	TCP	open	ajp13	Apache Jserv Protocol v1.3	nmap -sV
8180/tcp	TCP	open	http	Apache Tomcat/Coyote JSP 1.1	nmap -sV
8787/tcp	TCP	open	drb	Ruby DRb RMI	nmap -sV
32901/tcp	TCP	open	mountd	RPC #100005	nmap -sV
38448/tcp	TCP	open	status	RPC #100024	nmap -sV
46947/tcp	TCP	open	nlockmgr	RPC #100021	nmap -sV

Port	Proto	State	Service	Version / Evidence	Command
59707/tcp	TCP	open	java-rmi	GNU Classpath grmiregistry	nmap -sV
53/udp	UDP	open	domain	ISC BIND 9.4.2	nmap -sU
68/udp	UDP	open filtered	dhcpc	DHCP Client	nmap -sU
69/udp	UDP	open filtered	tftp	Trivial FTP	nmap -sU
111/udp	UDP	open	rpcbind	RPC #100000	nmap -sU
137/udp	UDP	open	netbios-ns	NetBIOS Name Service	nmap -sU
138/udp	UDP	open filtered	netbios-dgm	NetBIOS Datagram	nmap -sU
2049/udp	UDP	open	nfs	NFS RPC	nmap -sU

8. Student Questions

8.1 What is reconnaissance?

Reconnaissance is the process of gathering information about a target system or network to identify potential vulnerabilities and attack vectors before launching an attack. It is the first phase of ethical hacking and penetration testing.

8.2 What is the difference between host discovery and port scanning?

Host discovery (ping sweep) identifies which IP addresses are active on a network, while port scanning checks which ports/services are open on discovered hosts. Host discovery is less intrusive and faster.

8.3 Explain open, closed and filtered.

State	Description
Open	A service is actively accepting connections/traffic on that port.
Closed	The port is reachable but no service is listening (host responded with RST).
Filtered	Nmap cannot determine the state because a firewall or network filter is blocking probes.

8.4 What does -sV do?

The -sV flag enables service/version detection. Nmap sends probes to open ports to identify the specific software product and version number running behind each service.

8.5 What does -O do?

The -O flag enables operating system detection through TCP/IP stack fingerprinting. Nmap analyzes network responses to determine the target's operating system.

8.6 Explain what -A combines and why it is noisier.

The -A flag combines:

- OS detection (-O)
- Version detection (-sV)
- Default NSE scripts (-sC)
- Traceroute

It's noisier because it sends multiple probes and makes more network connections, generating more traffic.

8.7 What does -p- mean?

The -p- flag means scan all 65,535 TCP ports, from port 1 through port 65535, rather than the default 1,000 most common ports.

8.8 Why can UDP scans take longer?

UDP scans take longer because UDP is connectionless. When a UDP port is closed, many systems send an ICMP port unreachable message, but when a port is open or filtered, there's no response. Nmap must wait for timeouts, making the scan slower.

8.9 What does -sC do?

The -sC flag runs the default set of Nmap NSE (Nmap Scripting Engine) scripts for service enumeration, providing detailed information about services running on open ports.

8.10 What information did the HTTP title/headers scripts reveal?

- HTTP Title: "Metasploitable2 - Linux"
- HTTP Headers: Server: Apache/2.2.8 (Ubuntu) DAV/2, X-Powered-By: PHP/5.2.4-2ubuntu5.10

8.11 What did SMB/SSH/FTP enumeration add?

Service	Findings
SMB	NT LM 0.12 (SMBv1) protocol support (dangerous and default), Samba 3.0.20-Debian, Workgroup name
SSH	DSA (1024-bit) and RSA (2048-bit) host keys
FTP	vsftpd 2.3.4 banner, anonymous FTP login allowed

8.12 What is the difference between Nmap service detection and WhatWeb fingerprinting?

Nmap Service Detection	WhatWeb Fingerprinting
Identifies network services and their versions at the protocol level	Specifically fingerprints web technologies
Works across all ports	Focuses on HTTP/HTTPS services
Identifies services like SSH, FTP, MySQL	Identifies HTTP servers, frameworks, CMS, JavaScript libraries

8.13 How did WhatWeb levels 1, 3 and 4 differ?

Level	Description	Speed
Level 1	Minimal requests, basic fingerprinting	Fast
Level 3	More requests for better identification	Medium
Level 4	Heavy requests, substantially more probes	Very Slow

8.14 Why must aggressive scanning/fingerprinting remain inside the authorised lab?

Aggressive scanning can:

- Cause network disruption
- Trigger security alerts
- Be considered unauthorized/illegal against production systems
- Potentially be classified as a cyber attack

9. Command Cheat Sheet

Purpose	Command
Host Discovery	<code>nmap -sn 192.168.19.0/24</code>
Default Scan	<code>nmap 192.168.19.133</code>
Service Versions	<code>nmap -sV 192.168.19.133</code>
OS Detection	<code>sudo nmap -O 192.168.19.133</code>
Aggressive Scan	<code>sudo nmap -A 192.168.19.133</code>
Full TCP Scan	<code>sudo nmap -p- 192.168.19.133</code>
UDP Scan	<code>sudo nmap -sU 192.168.19.133</code>
Default NSE Scripts	<code>nmap -sC -sV 192.168.19.133</code>
WhatWeb Basic	<code>whatweb http://192.168.19.133</code>
WhatWeb Level 3	<code>whatweb -a 3 -v http://192.168.19.133</code>

10. Conclusion

This lab successfully demonstrated network reconnaissance techniques using Nmap and WhatWeb against the Metasploitable 2 target. The following objectives were achieved:

- ✓ Kali and Metasploitable IP addresses were identified
- ✓ Connectivity between VMs was confirmed
- ✓ Host discovery identified live systems on the network
- ✓ Default TCP scans revealed 23 open ports
- ✓ Service version detection identified specific software versions
- ✓ OS detection identified Linux kernel 2.6
- ✓ Aggressive scanning provided comprehensive service information
- ✓ Full TCP scans revealed all 31 open ports
- ✓ UDP scans identified 7 UDP services
- ✓ NSE scripts provided detailed service enumeration
- ✓ WhatWeb fingerprinted web technologies
- ✓ A complete service inventory was created

All scans were performed within the authorised lab environment, and findings were documented for evidence and reporting purposes.

11. Completion Checklist

Task	Status
I used only the authorised target	✓
I completed all required Nmap scan types	✓
I can explain each important option	✓
I completed safe NSE enumeration	✓
I compared WhatWeb aggression levels	✓
I built a final service inventory	✓
I answered all questions in my own words	✓